ELECTRICAL—AIR BAG LAMP ILLUMINATED AND/OR DTC'S B2295, B2296 OR B2477 STORED IN MEMORY	Article No. 02-11-5

FORD: 2001-2002 WINDSTAR

ISSUE

Some vehicles may exhibit an intermittent air bag lamp illuminated and/or Diagnostic Trouble Codes (DTCs) B2295, B2296 or B2477. These DTCs may be caused by loose/not mated connections, trapped/chafed wires beneath the seat(s) or at the Restraints Control Module (RCM), sensors not properly grounded, incorrect RCM version, and/or non functioning front or side crash sensors.

ACTION

Verify Diagnostic Trouble Codes (DTCs) B2295, B2296 or B2477. For general help with all restraint component diagnostics, refer to the following Service Procedure and the attached Quick Sheets (Figures 3 through 7) for details.

SERVICE PROCEDURE

Service is to include at least one if not all of the following elements executed in the order listed:

1. Workshop Manual deactivation procedure. Workshop Manual Section 501-20B.
2. "Quick Checks" as detailed in the diagnostic Quick Sheets (Figures 3 through 7) "Remaining Actions to take".
3. Workshop Manual pinpoint tests. Check "Obscure Causes" as detailed in the diagnostic Quick Sheets under "Remaining Actions to Take".
4. Check for loose/not mated connections, trapped/chafed wires beneath the seat(s) or at the RCM, sensors not properly grounded, incorrect RCM version, and/or non functioning front or side crash sensors.

The procedures below apply to DTC's B2295, B2296 and B2477

1. Perform the Workshop Manual deactivation procedure. Workshop Manual Section 501-20B.
2. Use NGS or WDS to retrieve the DTC codes from the RCM. Be sure to also press the NGS "FLAG" key if available, after retrieving the DTC's. This will retrieve the PID's that will help you determine root cause in diagnosing each failed circuit. If the light is on when diagnosing the vehicle determine the flash code. Refer to the flash code guide in the Workshop Manual or the attached Quick Sheets (Figures 3 through 7). If the flash codes are 4-2 (front), 4-3 (driver), or 4-4 (passenger) with DTC B2296 active, this indicates conditions with the front, drivers side, or passenger side crash sensor circuit(s). If the flash codes are 2-2 (drivers) or 2-3 (passenger) with DTC B2295 active, this indicates conditions with the side air bag circuit(s). Be sure to note all active flash codes, DTC's and PID responses.
3. Perform any "Quick Checks" as detailed in the diagnostic Quick Sheets "Remaining Actions to Take".
4. Perform Workshop Manual pinpoint tests.

NOTE

THE WORKSHOP MANUAL PINPOINT TESTS ARE VERY EFFECTIVE AT DETERMINING ROOT CAUSE OF HARD (CURRENTLY ACTIVE) CIRCUIT FAULTS. IF THE CIRCUIT FAILURE YOU ARE INVESTIGATING IS NOT A HARD FAILURE, BUT INTERMITTENT, YOU WILL HAVE TO PERFORM THE PINPOINT TESTS IN COMBINATION WITH SOME HARNESS FLEXING AND CIRCUIT OVERLAYS TO DETERMINE ROOT CAUSE.

5. Check for any "Obscure Causes" as detailed in the diagnostic Quick Sheets under "Remaining Actions to Take" (Figures 3 through 7).
6. Check for loose/not mated connections. Check for trapped/chafed wires beneath the seat(s) or at the RCM (Figures 1 and 2 for locations to check). Check for sensors not properly grounded and for incorrect RCM version.

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7. If no issues with the wiring and/or mounting are found after all six procedures above have been completed in detail, continue with the procedure.

DTC B2296

Replace the necessary crash sensor 2F2Z-14B004-AA for the front sensor or 2F2Z-14B345-AA for the side sensor.

DTC B2295

1. Determine which side air bag circuit is at fault, drivers or passengers.
2. If the vehicle is not equipped with side air bags measure the resistance of the delete resistor. Resistance should be 6.8 ohms +/- .5 ohms.
3. If the vehicle is equipped with side air bags, install the side air bag simulator and retest the vehicle. If the problem is corrected, then the condition is within the seat.
4. Inspect the following areas in the seat for cut wires:
   - Under the seat
   - Side air bag wiring at the back of the seat back (Figures 1 and 2)
5. If the wiring for the side air bag is cut replace the side air bag:
   - Drivers "Side" Air Bag 2F2Z-16611D11-AA
   - Passengers "Side" Air Bag 1F2Z-16611D10-AA

DTC 2477

2001 Windstars have two different types of RCMs released for service based on the build date. Vehicles built prior to April 2001 were not equipped with the Occupant Classification Sensor (OCS) on the passengers seat. Vehicles build after April 1, 2001 were equipped with the OCS. Different RCMs are required to support the vehicles with the OCSs. If the RCM for the OCS is installed in a vehicle without the OCS system, an OCS communication fault will occur, DTC B2290. If the RCM not equipped with the OCS is installed in a vehicle equipped with the OCS a configuration fault will occur DTC B2477. The following are the correct RCM part numbers:

1. If the vehicle is not equipped with the OCS (built prior to April 1, 2001).
   a. Vehicle equipped with side air bag 1F2Z-14B321-DB.
   b. Vehicle without side air bag 1F2Z-14B321-CB.
2. If the vehicle is equipped with the OCS (built after April 1, 2001).
   a. Vehicle equipped with side air bag 1F2Z-14B321-BA.
   b. Vehicle without side air bag 1F2Z-14B321-AA.

PART NUMBER	PART NAME
2F2Z-14B345-AA	Side Crash Sensor
2F2Z-14B004-AA	Front Crash Sensor
1F2Z-14B321-CB	RCM Module - w/o Side Air Bag - Built Before 4/1/2001
1F2Z-14B321-DB	RCM Module - with Side Air Bag - Built Before 4/1/2001
1F2Z-14B321-AA	RCM Module - w/o Side Air Bag - Built After 4/1/2001
1F2Z-14B321-BA	RCM Module - with Side Air Bag - Built After 4/1/2001
1F2Z-16611D10-AA	Passenger "Side" Air Bag
2F2Z-16611D11-AA	Driver "Side" Air Bag

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 104000, 206000

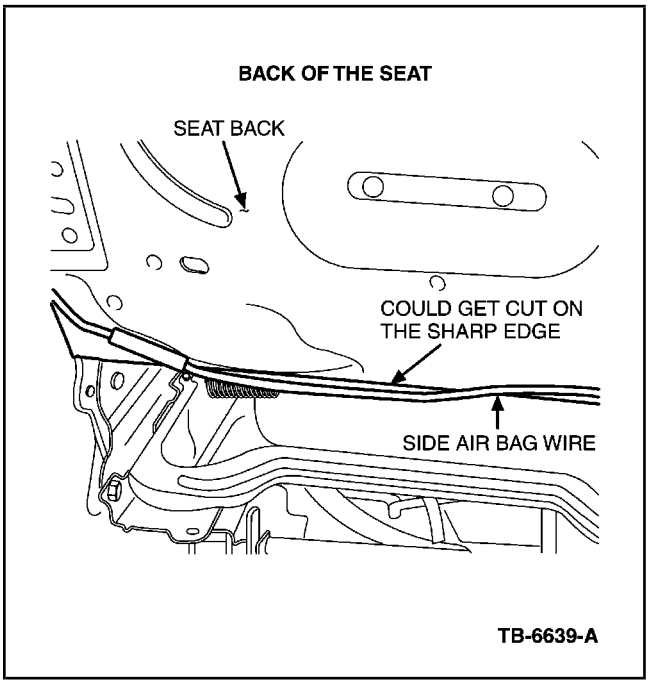


Figure 1 - Article 02-11-5

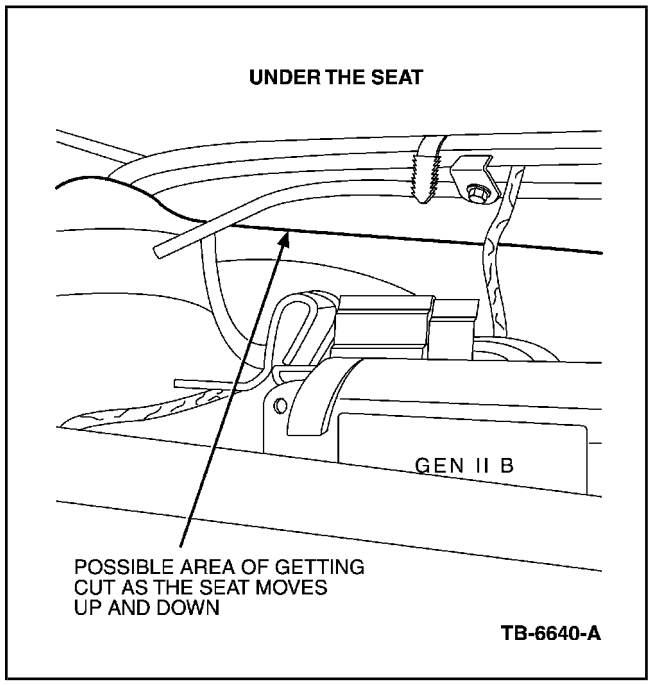
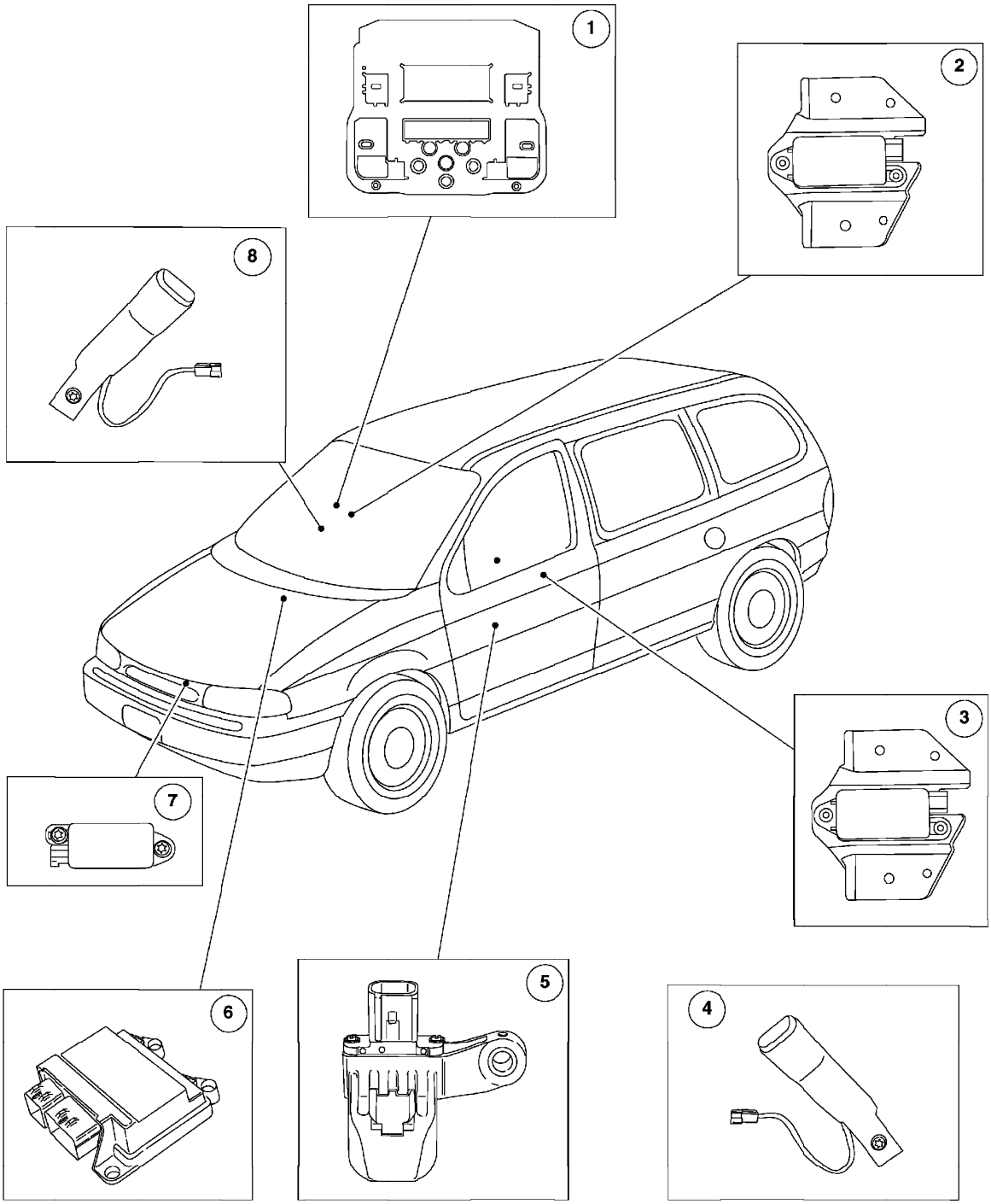


Figure 2 - Article 02-11-5

WINDSTAR 2001/2002 MODEL YEAR RESTRAINTS COMPONENTS DIAGNOSTIC "QUICK SHEET"



NOTE 1: Of all of the system faults currently active (On Demand), only the highest priority fault is flashed.

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Figure 3 - Article 02-11-5

Figure 4 - Article 02-11-5

WINDSTAR 2001/2002 MODEL YEAR RESTRAINTS COMPONENTS DIAGNOSTIC "QUICK SHEET" (See Component Illustrations in Figure 3)

Restraints Component Name	Part Number	Diagnostic Trouble Code (DTC)	Lamp Flash Code (LFC)	LFC Priority (Note 1)	Fault Description	"Detailed Diagnostics" Parameter Identifier PID (NGS "FLAG" Key)	1st Action To Take: System Deactivation Procedure (see Service Manual) Remaining Actions to Take Below: 1. Quick Checks 2. Service Manual Pinpoint Test 3. Obscure Causes
(1) Passenger Occupant Classification Sensor **(OCS)**		B2290	16	19	Occupant Classification Sensor (Seat Weight) Fault	**On Demand: 5959 Continuous: 5952 Bit Mapped Detail:** OCS Internal Module Fault, Communications Fault, Calibration Fault, Sensing Element Fault.	**1.** Check connections under passenger seat. **2.** Pinpoint Test Defined. **3.** None known.
		B2477	5-4	7	Incorrect Vehicle Configuration	**None**	**1.** If passsenger seat contains OCS, the RCM must detect the OCS. There are two RCM versions (OCS and non-OCS). Confirm RCM version matches passenger seat type. **2.** No Pinpoint Test. **3.** None known.
(2) Passenger Side Impact Sensor (Passenger SCS)	14B345	B2296	4-4	10	Passenger Side Impact Sensor Circuit Failure--> PID Diagnostic Detail Available	**On Demand: 5965 Continuous: 5958 Bit Mapped Detail:** Sensor Mount/ Communications Fault, Sensor Internal Fault.	**1.** Check sensor under seat for secure mounting and harness issues. **2.** Pinpoint Test Defined. **3.** None known.
(3) Driver Side Impact Sensor (Driver SCS)	14B345	B2296	4-3	9	Driver Side Impact Sensor Circuit Failure--> PID Diagnostic Detail Available	**On Demand: 5965 Continuous: 5958 Bit Mapped Detail:** Sensor Mount/ Communications Fault, Sensor Internal Fault.	**1.** Check sensor under seat for secure mounting and harness issues. **2.** Pinpoint Test Defined. **3.** None known.
(4) Driver Belt Buckle Sensor	1661203	B2432	5-1	11	Driver Seat Belt Buckle Sensor Circuit Open	**None**	**1.** Driver seatbelt sensor and harness. **(NOTE: The sensor CANNOT be tested as if it were a simple switch.** Install known good after harnesss check. **2.** Pinpoint Test Defined. **3.** Check C305 and C304 under seat. Check ground connection left side kick panel.
		B2433	5-1	11	Driver Seat Belt Buckle Sensor Circuit Short to Battery	**None**	**1.** Driver seatbelt sensor and harness. **(NOTE: The sensor CANNOT be tested as if it were a simple switch.** Install known good after harnesss check. **2.** Pinpoint Test Defined. **3.** None known.
		B2434	5-1	11	Driver Seat Belt Buckle Sensor Circuit Short to Ground	**None**	**1.** Driver seatbelt sensor and harness. **(NOTE: The sensor CANNOT be tested as if it were a simple switch.** Install known good after harnesss check. **2.** Pinpoint Test Defined. **3.** None known.
		B2435	5-1	11	Driver Seat Belt Buckle Sensor Circuit Current out of Range	**None**	**1.** Driver seatbelt sensor and harness. **(NOTE: The sensor CANNOT be tested as if it were a simple switch.** Install known good after harnesss check. **2.** Pinpoint Test Defined. **3.** Check C305 and C304 under seat. Check ground connection left side kick panel.
(5) Driver Seat Track Sensor	14B416	C1946	4-9	21	Driver Seat Track Sensor Circuit Open	**None**	**1.** Driver seat track sensor and harness. **(NOTE: The sensor CANNOT be tested as if it were a simple switch.** Install known good after harnesss check. **2.** Pinpoint Test Defined. **3.** Check C305 and C398 under seat. Check ground connection left side kick panel.
		C1982	4-9	21	Driver Seat Track Sensor Circuit Short to Battery	**None**	**1.** Driver seat track sensor and harness. **(NOTE: The sensor CANNOT be tested as if it were a simple switch.** Install known good after harnesss check. **2.** Pinpoint Test Defined. **3.** Check C305 and C398 under seat. Check ground connection left side kick panel.
		C1947	4-9	21	Driver Seat Track Sensor Circuit Short to Ground	**None**	**1.** Driver seat track sensor and harness. **(NOTE: The sensor CANNOT be tested as if it were a simple switch.** Install known good after harnesss check. **2.** Pinpoint Test Defined. **3.** Check C305 and C398 under seat. Check ground connection left side kick panel
		C1948	4-9	21	Driver Seat Track Sensor Circuit Current out of Range	**None**	**1.** Driver seat track sensor and harness. **(NOTE: The sensor CANNOT be tested as if it were a simple switch.** Install known good after harnesss check. **2.** Pinpoint Test Defined. **3.** Check C305 and C398 under seat. Check ground connection left side kick panel.

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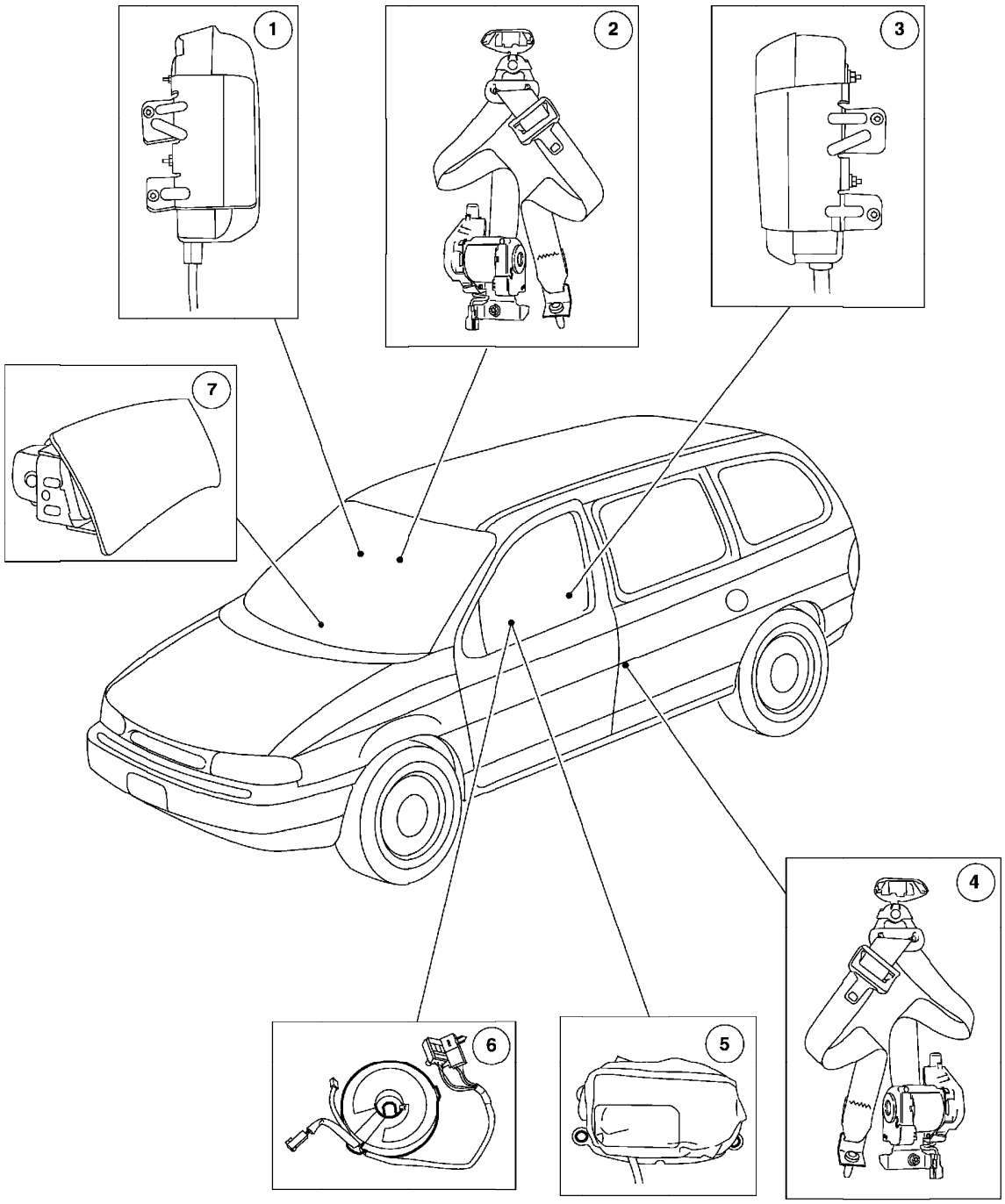
Figure 5 - Article 02-11-5

WINDSTAR 2001/2002 MODEL YEAR RESTRAINTS COMPONENTS DIAGNOSTIC "QUICK SHEET" (See Component Illustrations in Figure 3)

Restraints Component Name	Part Number	Diagnostic Trouble Code (DTC)	Lamp Flash Code (LFC)	LFC Priority (Note 1)	Fault Description	"Detailed Diagnostics" Parameter Identifier PID (NGS "FLAG" Key)	1st Action To Take: System Deactivation Procedure (see Service Manual) Remaining Actions to Take Below:
(6) Restraints Control Module **(RCM)**	14B321	B1231	1-3	4	RCM Crash Data Memory Full	**None**	**1.** Call Hotline. Get replacement permission. Return RCM to Ford for analysis. **2.** No Pinpoint Test. **3.** None known.
		B1342	1-2	3	RCM Internal Failure	**None**	**1.** If DTC B2296 also active, do Pinpoint for B2296. **2.** No other Pinpoint. **3.** A: Check for any combination of Driver/Passenger buckle and/or seat track sensor feed lines shorted together. B: Check for a leakage path (harness short, connector water etc.) in any of the Front or Side Impact Sensors. C: Occurs if this is the RCM's 5th crash.
		B1921	1-4	5	RCM Mounting Failure	**None**	**1.** Check RCM for secure mounting. **2.** Pinpoint Test Defined. **3.** None known.
		C1414	1-5	6	Incorrect Vehicle Identification	**PID: 5968** **Bit Mapped Detail:** RCM Version Conflicts with Version of OCS, Hard Wire VID, FCS, Driver SCS, Passenger SCS	**1.** Check RCM, OCS, FCS, Driver SCS, Passenger SCS for correct version. **2.** Pinpoint Test Defined. **3.** Connector (C292) under driver side IP (behind knee bolster). Ground connection left side kick panel. Check connections under passenger seat.
		N/A	N/A	N/A	No Communication with the RCM	**None**	**1.** Check Scan Tool, Data Link Connector (DLC), Harness. **2.** Pinpoint Test Defined. **3.** None known.
		N/A	Continuous Solid Lamp	2	RCM disconnected, or RCM inoperative or Loss of Battery feed, or Loss of Signal Ground	**None**	**1.** Check connections, fuse, harness, battery and ground at the RCM. **2.** Pinpoint Test Defined. **3.** Left kick panel ground connection.
		B1318	Continuous Solid Lamp	2	Battery Voltage too Low	**None**	**1.** Check if battery voltage at the RCM is < 9Vdc. **2.** Procedure Defined. **3.** None known.
		B1317	Continuous Solid Lamp	2	Battery Voltage too High	**None**	**1.** Check if battery voltage at the RCM is > 16Vdc. **2.** Procedure Defined. **3.** None known.
		Any Combo Of: B2290 B2295 B2432 B2436 C1946	5 Hertz Flashing Forever	23	B and E Plant Mode	**None**	**1.** Check connections for all seat sensors (OCS, buckle, seat track). **2.** No Pinpoint Test. **3.** None known.
(7) Front Impact Severity Sensor (FCS)	14B006	B2296	4-2	8	Front Impact Severity Sensor Circuit Failure--> PID Diagnostic Detail Available	**On Demand: 5965** **Continuous: 5958** **Bit Mapped Detail:** Sensor Mount/ Communications Fault, Sensor Internal Fault.	**1.** Check sensor of Rad support for secure mounting and harness issues. Check C180 behind left IP. **2.** Pinpoint Test Defined. **3.** None known.
(8) Passenger Belt Buckle Sensor	166120	B2436	5-2	12	Passenger Seat Belt Buckle Sensor Circuit Open	**None**	**1.** Passenger seat belt sensor and harness. **(NOTE: The sensor CANNOT be tested as if it were a simple switch.** Install known good after harness check. **2.** Pinpoint Test Defined. **3.** Check C366 and C399 under seat. Check ground connection right side kick panel.
		B2437	5-2	12	Passenger Seat Belt Buckle Sensor Circuit Short to Battery	**None**	**1.** Passenger seat belt sensor and harness. **(NOTE: The sensor CANNOT be tested as if it were a simple switch.** Install known good after harness check. **2.** Pinpoint Test Defined. **3.** None known.
		B2438	5-2	12	Passenger Seat Belt Buckle Sensor Circuit Short to Ground	**None**	**1.** Passenger seat belt sensor and harness. **(NOTE: The sensor CANNOT be tested as if it were a simple switch.** Install known good after harness check. **2.** Pinpoint Test Defined. **3.** None known.
		B2439	5-2	12	Passenger Seat Belt Buckle Sensor Circuit Current out of Range	**None**	**1.** Passenger seat belt sensor and harness. **(NOTE: The sensor CANNOT be tested as if it were a simple switch.** Install known good after harness check. **2.** Pinpoint Test Defined. **3.** Check C366 and C399 under seat. Check ground connection right side kick panel.

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WINDSTAR 2001/2002 MODEL YEAR RESTRAINTS COMPONENTS DIAGNOSTIC "QUICK SHEET"



NOTE 1: Of all of the system faults currently active (On Demand), only the highest priority fault is flashed.

NOTE 2: If a second fault (other than the lamp fault) is currently active (On Demand), then a secondary tone will be generated.

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Figure 6 - Article 02-11-5

Figure 7 - Article 02-11-5

WINDSTAR 2001/2002 MODEL YEAR RESTRAINTS COMPONENTS DIAGNOSTIC "QUICK SHEET" (See Component Illustrations in Figure 6)

Diagnostic Component Name	Part Number	Diagnostic Trouble Code (DTC)	Lamp Flash Code (LFC)	LFC Priority (Note 1)	Fault Description	"Detailed" Diagnostics" Parameter Identifier PID (NGS "FLAG" Key)	1st Action to Take System Deactivation Procedure (see Service Manual) Remaining Actions to Take (below): 1. Quick checks 2. Service Manual Pinpoint Test 3. Obscure Cases
(1) Passenger Seat - Side Airbag Module	611D10	B2295	2-3	18	Passenger Side Airbag Circuit Failure => PID Diagnostic Detail Available	**On Demand: 5964** **Continuous: 5957** **Bit Mapped Detail:** Circuit Short to Battery, Short to Ground, Open, Resistance Low	**1.** Yellow connectors under seats C334 C333. Seat to airbag jumpers C330 C331. **2.** Pinpoint Test defined. **3.** None known.
(2) Passenger Belt Retractor & Pretensioner	611B08	B2292	3-4	16	Passenger Pretensioner Circuit Failure => PID Diagnostic Detail Available	**On Demand: 5964** **Continuous: 5957** **Bit Mapped Detail:** Circuit Short to Battery, Short to Ground, Open, Resistance Low	**1.** Passenger B-pillar connector. **2.** Pinpoint Test defined. **3.** None known.
(3) Driver Seat - Side Airbag Module	611D11	B2295	2-2	17	Driver Side Airbag Circuit Failure => PID Diagnostic Detail Available	**On Demand: 5964** **Continuous: 5957** **Bit Mapped Detail:** Circuit Short to Battery, Short to Ground, Open, Resistance Low	**1.** Yellow connectors under seats C334 C333. Seat to airbag jumpers C330 C331. **2.** Pinpoint Test defined **3.** None known.
(4) Driver Belt Retractor & Pretensioner	611B09	B2292	3-3	15	Driver Pretensioner Circuit Failure => PID Diagnostic Detail Available	**On Demand: 5961** **Continuous: 5954** **Bit Mapped Detail:** Circuit Short to Battery, Short to Ground, Open, Resistance Low	**1.** Seatbelt Sensor and other B-pillar C385. **2.** Pinpoint Test defined **3.** None known.
(5) Driver Front Airbag Module	043B13	B2293	1-9	13	Driver Front Airbag Circuit Failure => PID Diagnostic Detail Available	**On Demand: 5962** **Continuous: 5955** **Bit Mapped Detail:** Circuit Short to Battery, Short to Ground, Open, Resistance Low	**1.** Check C296 Knee Bolster, C267, Top Steer Column, C268 Clockspring to bag. **2.** Pinpoint Test defined. **3.** None known.
(6) Clockspring	14A664	B2293	1-9	13	Driver Front Airbag Circuit Failure => PID Diagnostic Detail Available	**On Demand: 5962** **Continuous: 5955** **Bit Mapped Detail:** Circuit Short to Battery, Short to Ground, Open, Resistance Low	**1.** Check C296 Knee Bolster, C267, Top Steer Column, C268 Clockspring to bag. **2.** Pinpoint Test defined **3.** None known
(7) Passenger Front Airbag Module	044A74	B2293	2-1	14	Passenger Front Airbag Circuit Failure => PID Diagnostic Detail Available	**On Demand: 5962** **Continuous: 5955** **Bit Mapped Detail:** Circuit Short to Battery, Short to Ground, Open, Resistance Low	**1.** Check C269 behind glovebox. C296 Knee Bolster. **2.** Pinpoint Test defined **3.** None known.
(NOT PICTURED) Instrument Cluster		B1869	Note 2	1	Airbag Warning Indicator Lamp Circuit Open or Short to Ground	**None**	**1.** Check lamp fuses, lamp itself, C241. **2.** Pinpoint Test defined **3.** None known.
		B1870	Note 2	1	Airbag Warning Indicator Lamp Circuit Short to Battery	**None**	**1.** Check harness. **2.** Pinpoint Test defined **3.** None known.
		B1891	5-3	22	Airbag Tone Warning Indicator Circuit Short to Battery	**None**	**1.** Check harness. **2.** Pinpoint Test defined **3.** None known.
		B1892	5-3	22	Airbag Tone Warning Indicator Circuit Open or Short to Ground	**None**	**1.** Check harness for shorts to ground or open circuit, cluster fuse, C240. **2.** Pinpoint Test defined. **3.** None known.

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